

Global Tactical Asset Allocation

Updated Results and Real-Market Implementation Using Python and IBKR

Mohamed Gabriel, Alberto Pagani, Carlo Zarattini

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1 Introduction

This study revisits and expands on one of the most influential investment research papers of the past two decades: *A Quantitative Approach to Tactical Asset Allocation*, authored by Meb Faber in 2006 and published in *The Journal of Wealth Management* in 2007.

We begin with a concise overview of the original strategy, then present updated historical performance results using data through March 2025. Next, we explore how different rebalancing frequencies can affect outcomes, and propose a tranche-based approach to help practitioners reduce the impact of rebalance timing luck.

Finally, we show how investors can put this strategy into practice with a simple Python script that automates portfolio rebalancing through Interactive Brokers.

2 Global Tactical Asset Allocation

Global Tactical Asset Allocation (GTAA) is an active investment approach that dynamically adjusts a portfolio's exposure across a range of global assets, aiming to deliver a more favorable risk/reward profile than a purely passive benchmark.

Unlike strategic asset allocation, which maintains fixed asset weights over the long term, GTAA permits temporary shifts in asset exposure based on market conditions. For example, the strategy might increase equity exposure during bullish periods or move to cash during market downturns. These allocation decisions are driven entirely by quantitative, rules-based signals.

As outlined in Meb Faber's original paper, the strategy dynamically allocates capital across five total return indices:

- S&P 500 Index
- MSCI EAFE Index
- U.S. 10-Year Government Bonds
- Goldman Sachs Commodity Index (GSCI)
- National Association of Real Estate Investment Trusts (NAREIT)

For each index, a 200-day simple moving average is used as a trend-following signal. A *buy signal* is generated when the end-of-month closing price is above the 200-day average, while a *sell signal* is issued when the price falls below it. In the latter case, the corresponding allocation is shifted to cash. Signals are evaluated exclusively at month-end, and trades are executed immediately thereafter¹.

When a buy signal is present, the associated ETF receives a 20% allocation. Any remaining capital is allocated to 90-day U.S. Treasury bills.²

All index and price data are sourced from Bloomberg. This analysis excludes taxes, commissions, and slippage.

2.1 Updated Results

Meb Faber's original backtest covered the period from 1972 to the end of 2005. Over that *in-sample* window, the strategy delivered a Sharpe Ratio of 0.81, a compound annual growth rate (CAGR) of 11.7%, and a maximum drawdown (MDD) of 9.5%³.

¹For results incorporating a one-day execution lag after signal generation, please refer to Section 3.

²Treasury bill returns are obtained from the prof. Kenneth R. French website.

³This drawdown was calculated using monthly observations and may slightly underestimate the true maximum drawdown at daily

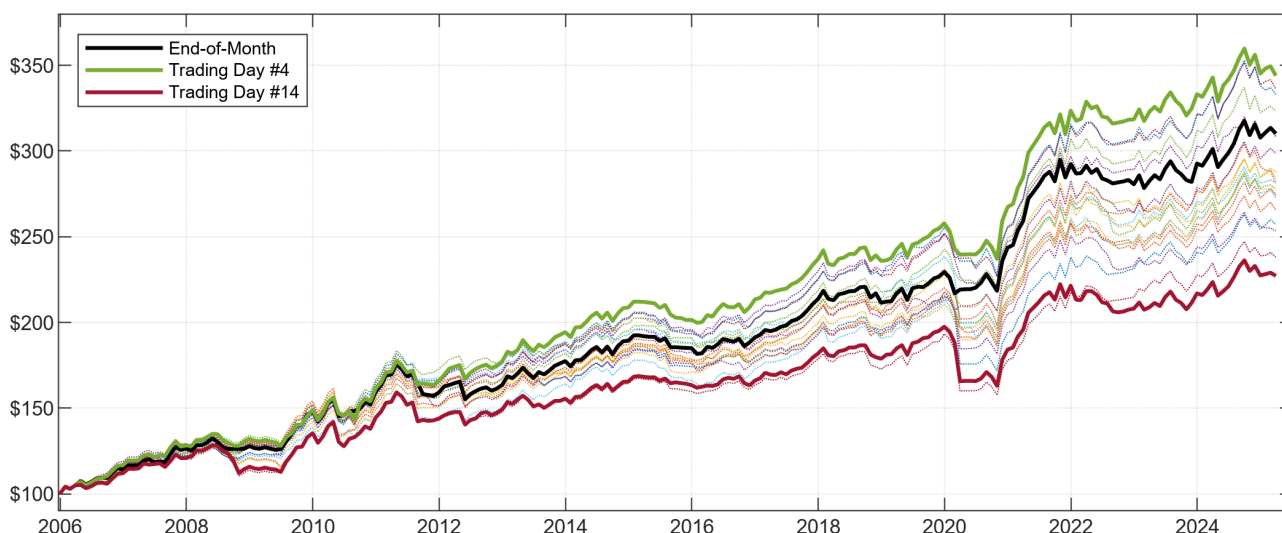


Figure 1: Simulation of twenty different GTAA portfolios, each differing by rebalancing date (i.e. n th trading day of the month). The best performing run, worst performing run, and the end of month rebalance implemented are highlighted in bold.

Our updated backtest covers the *out-of-sample* period from 2006 to March 2025. The results remain solid: a Sharpe Ratio of 0.68, a CAGR of 6.05%, and a maximum drawdown of 11.7%. These figures are especially impressive given the series of market shocks over the past two decades, including the Global Financial Crisis (2008–2009), the European Sovereign Debt Crisis (2011), the Taper Tantrum (2013), the China Crisis (2015), Volmageddon and the U.S.–China Trade War (2018), the COVID-19 pandemic (2020), and the inflation-driven selloffs of 2022.

2.2 Rebalance Timing Luck

However, one important aspect often overlooked in strategies that rebalance infrequently and rely on binary signals is rebalance timing luck. Put simply: how much do results change depending on *when* you rebalance?

To investigate this, we ran simulations for 21 different GTAA portfolios, each rebalancing on a different trading day of the month. Specifically, the portfolios compute signals and execute trades on the n -th trading day of each month, where n ranges from 0 to 20. For example, when $n = 0$, the strategy rebalances on the last trading day of the month, as in the original Faber approach. When $n = 1$, rebalancing occurs on the first trading day of the month. This pattern continues sequentially through the 20th trading day of the month.

As shown in Figure 1, there is significant dispersion in performance depending on the rebalancing date. The compound annual growth rate (CAGR) differs by nearly 220 basis points between the best-performing case (rebalancing on the 4th trading day) and the

worst (14th trading day). The maximum drawdown⁴ increases sharply from 10.75% to over 18%, while the Sharpe Ratio deteriorates from 1.02 to 0.72.

A straightforward, albeit foundational, approach to mitigating rebalance timing luck is to increase the rebalancing frequency. For instance, one could rebalance weekly on a predefined day. The simplicity of this solution lies in the fact that reducing the number of rebalancing trajectories (e.g., from 20 to 5) and narrowing the time lag across signals naturally leads to reduced dispersion in outcomes.

2.3 Weekly Rebalancing

We reran the simulation using a weekly rebalancing schedule, with signals and trades executed at the closing price on specific weekdays. Empirical results confirmed our intuition: outcome dispersion is significantly reduced. The CAGR spread between the best and worst rebalancing days shrank to under 130 basis points; excluding Monday (the weakest performer), the spread narrows further to below 50 basis points.

The main drawback of increasing rebalancing frequency, especially when using binary signals, is the risk of underperformance in mean-reverting market environments. For example, consider a scenario in which signals alternate between 1 and 0 over five consecutive Mondays. A system that rebalances weekly on Mondays would repeatedly enter and exit positions, incurring losses from adverse price reversals and elevated transaction costs. By contrast, a monthly rebalancing approach would remain insulated from this short-term noise, keeping positions unchanged throughout the month.

frequency.

⁴Calculated using daily data.

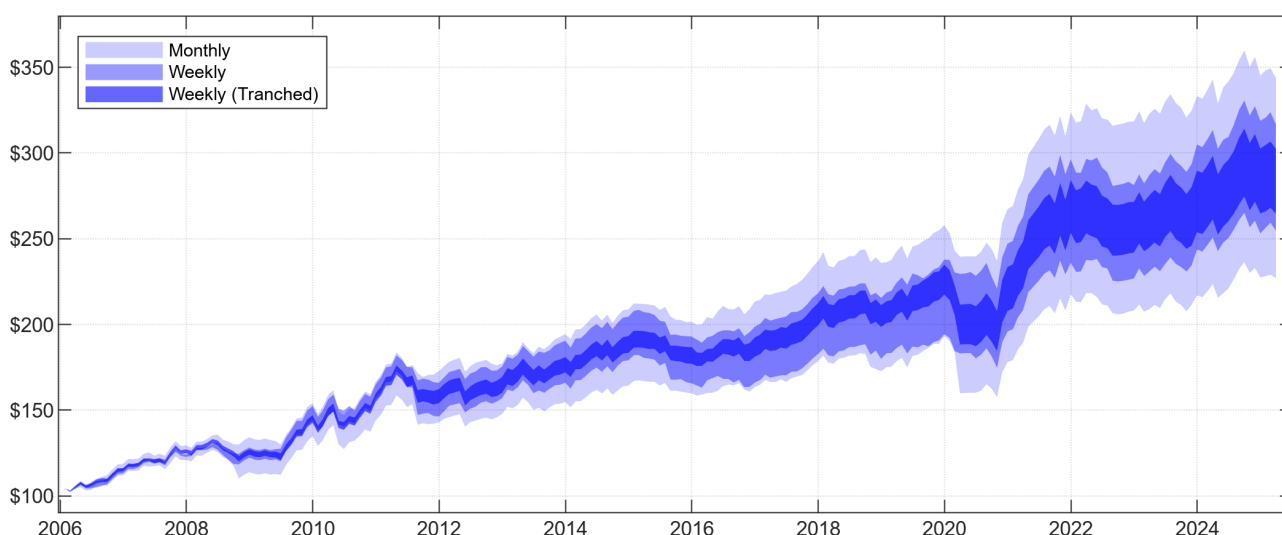


Figure 2: Dispersion of portfolio outcomes under different rebalancing schemes for the Global Tactical Asset Allocation (GTAA) strategy. The shaded regions represent the range between the best and worst performing rebalancing dates for each approach.

To address this challenge and move toward a more robust real-world implementation, we propose a *tranched rebalancing mechanism*. The portfolio is conceptually split into four sub-portfolios (or “tranches”), each rebalanced on a staggered four-week cycle.

In practice, this means we still rebalance weekly, but only one-quarter of the portfolio is adjusted at a time. This more refined framework reduces total turnover and simultaneously mitigates rebalance timing risk.

Figure 2 illustrates the range of possible outcomes for three implementations: the original monthly approach, weekly rebalancing, and the proposed tranched version. Notably, by distributing execution across four staggered tranches, the impact of poor timing is significantly reduced. The dispersion in compound annual growth rates (CAGR) across the tranched portfolios narrows to just 63 basis points, compared to 130 basis points observed under standard weekly rebalancing.

In Table 1, we present the main summary statistics for the weekly rebalanced strategy compared to the tranched implementation. The most relevant column is the *Turnover* (expressed in annual terms), which may initially appear elevated. This is largely due to the inclusion of cash transactions in the calculation. For example, when the system generates a sell signal for the S&P 500, it executes both a sell order for the S&P 500 and a simultaneous buy order of equivalent notional value in Treasury bills. This two-sided activity significantly increases the reported turnover.

By adopting a tranched approach, we were able to reduce annual turnover by nearly half. This improvement is not just theoretical; in practical terms,

Table 1: Weekly rebalancing vs. tranched: performance and turnover

Day	Tranched	CAGR	SR	MDD	Turnover
Mon	No	5.0%	0.70	15.7%	743%
Mon	Yes	5.2%	0.71	16.3%	398%
Tue	No	6.2%	0.86	11.5%	743%
Tue	Yes	5.9%	0.81	12.7%	385%
Wed	No	5.9%	0.83	10.4%	738%
Wed	Yes	5.7%	0.78	15.0%	386%
Thu	No	5.6%	0.79	10.5%	776%
Thu	Yes	5.6%	0.78	13.4%	387%
Fri	No	5.8%	0.82	13.8%	746%
Fri	Yes	5.6%	0.78	12.9%	391%

a reduction of 350% in turnover could translate into substantial cost savings. Assuming transaction costs of 10 basis points, this would imply a CAGR saving of approximately 35 basis points per year, which is highly relevant for long-term real-world implementation.

Even though we do not observe a consistent improvement in risk-adjusted returns between the tranched and non-tranched portfolios, this is not a concern. The objective of tranching is not to maximize performance, but rather to reduce exposure to rebalance timing luck, narrow the dispersion of portfolio outcomes, and lower portfolio turnover.

While the weekly tranched portfolio represents a more sound implementation, investors aiming to replicate it should remain mindful of their portfolio size. Because execution is distributed across multiple tranches, individual trade sizes may become relatively small. For smaller portfolios, this can lead to deviations from the theoretical allocation due to practical limitations, such as the inability to trade fractional shares, or the disproportionate impact of commissions on small transactions. For further discussion on these implementation

frictions, we refer readers to our paper “Does Trend Following Still Work on Stocks?” (2025b).

2.4 Signal Lag and Transaction Costs

To better reflect real-world conditions, we incorporate two key market frictions: signal lag and transaction costs. Specifically, after arbitrarily selecting one of the tranced portfolios (the one rebalancing on Wednesdays), we introduce a one-day delay between signal generation and trade execution. In this configuration, portfolio weights are calculated using Tuesday’s closing prices, while trades are executed at the close on Wednesday. In addition, we account for transaction costs by applying a fee of 10 basis points to all transacted notional amounts, a conservative estimate given the high liquidity of the instruments involved in the strategy.

Figure 3 plots the results. The first observation is that introducing a one-day lag between signal generation and execution does not negatively impact overall returns; in fact, it results in a slight improvement. The more substantial effect comes from the inclusion of transaction costs, which reduce performance by approximately 41 basis points.

While practitioners may adopt various portfolio management techniques to further reduce turnover, such as implementing rebalance thresholds, this is beyond the scope of the present study. For those interested, we refer readers to our previous papers, “Does Trend Following Still Work on Stocks?” (2025b) and “Catching Crypto Trends: A Tactical Approach for Bitcoin and Altcoins” (2025a), where these methods are examined in more detail. Additional practical consider-

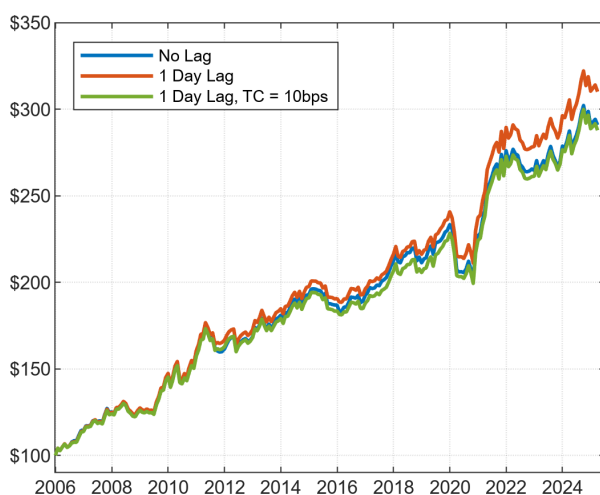


Figure 3: Impact of execution lag and transaction costs on portfolio performance for the GTAA strategy. Blue line highlights the baseline scenario with no lag or costs, the orange line incorporates a one-day signal execution lag, while the green line includes both a one-day lag and transaction costs of 10bps.

ations related to signal execution, transaction costs, and real-world portfolio constraints are discussed by Meb Faber in “A Quantitative Approach to Tactical Asset Allocation Revisited 10 Years Later” (2017).

3 Practical Implementation

In this section, we show how to implement the original Global Tactical Asset Allocation (GTAA) strategy introduced by Faber (2007) using Python. The full implementation is available as a Jupyter Notebook on Google Colab (link⁵). It is designed to be both educational and ready for real-world deployment, covering everything from data retrieval to trade execution.

Specifically, the notebook includes:

- Automatic data download from Polygon.io
- Signal generation using a trend-following model
- Trade execution via Interactive Brokers (IBKR), either manually (semi-automated) or programmatically (fully automated).

To simulate realistic trading conditions, all signals are generated using data available up to the previous trading day. Trades are then executed at the market close of the current day.

In line with the GTAA model introduced earlier, we focus on six highly liquid ETFs, as listed in Table 2.

Table 2: Benchmarks and corresponding ETFs used in the GTAA model

Benchmark	ETF Name	Ticker
S&P500	SPDR S&P 500 ETF Trust	SPY
MSCI EAFE	iShares MSCI EAFE ETF	EFA
US 10-Year	iShares 7-10 Year Treasury ETF	IEF
GSCI	Invesco DB Commodity Index ETF	DBC
NAREIT	Vanguard Real Estate ETF	VNQ
T-Bill	iShares Short Treasury Bond ETF	SHV

3.1 Interactive Portfolio Setup

The first step in the automatic rebalancing process is to define the amount of capital to be allocated to the GTAA strategy, along with the current holdings (if any) in the related ETFs.

This process is streamlined using interactive widgets that collect user inputs:

- Enter the amount of equity (AUM) to allocate to GTAA
- Specify the number of shares currently held in each ETF⁶

⁵<https://bit.ly/FaberIBKR>

⁶Only include holdings that belong to the GTAA strategy.

Figure 4 shows the interactive interface where users input their AUM and ETF positions.

Inputs submitted. Ready to fetch data and compute signals.
AUM: \$100,000.00
SPY: 100.0 shares
EFA: 100.0 shares
IEF: 100.0 shares
VNQ: 100.0 shares
DBC: 100.0 shares
SHV: 0.0 shares

Figure 4: Interface for entering AUM and ETF holdings.

3.2 Signal Generation

The signal generation process includes the following steps:

1. Download one year of daily adjusted closing prices for each ETF⁷
2. Compute the 200-day SMA for each ETF using the formula:

$$SMA_{i,t} = \frac{1}{200} \sum_{k=0}^{199} P_{i,t-k}$$

where $P_{i,t-k}$ is the adjusted closing price of ETF i , k days before yesterday (t).

3. Generate a binary trend-following signal (excluding SHV):

$$Signal_{i,t} = \begin{cases} 1 & \text{if } P_{i,t} > SMA_{i,t} \\ 0 & \text{otherwise} \end{cases}$$

4. ETFs with a positive signal are each assigned a fixed weight of 20%. ETFs with a signal of 0 receive no allocation.

3.3 Rebalancing Logic

Once signals and target allocations are defined, the notebook calculates the number of shares to hold in each ETF and determines the trades needed to adjust the portfolio accordingly. The rebalancing logic follows these steps:

1. **Calculate Target Shares.** For each ETF with a positive signal, the algorithm calculates the target number of shares to hold based on the current AUM and ETF price:

$$Shares_i = \left\lfloor \frac{AUM \cdot w_i}{P_{last,i}} \right\rfloor$$

where:

⁷It is recommended to use adjusted total return prices, which account for dividends and stock splits. This ensures that signals are not distorted by corporate actions that do not impact the stock's underlying economic value.

- AUM is the amount allocated to the GTAA strategy
- w_i is the weight assigned to ETF i (20% if the signal = 1, otherwise 0%)
- $P_{last,i}$ is the most recent price of ETF i .

If the ETF is not in an uptrend (signal = 0), its target weight and shares are both set to zero.

2. **Calculate Trade Required.** The difference between the current holdings and the target holdings determines how many shares to buy or sell:

$$\Delta Shares_i = Shares_i - Shares_{user,i}$$

Where $\Delta Shares_i$ is the quantity to trade for ETF i .

3. **Cash Allocation to SHV.** Any capital not allocated to trend-positive ETFs is invested in SHV, a short-term U.S. Treasury ETF. SHV is always considered to have a signal of 1, meaning it can receive any unallocated capital. The number of SHV shares is calculated based on the residual AUM after allocating to the other ETFs.

4. **Summary Table.** Once the rebalancing logic is executed, the notebook produces a summary table as shown in Table 3.

Table 3: Example trade summary output from the notebook

Ticker	Signal	Price	Current	Shares	$\Delta Shares$
SPY	0	527.32	150	0	-150
EFA	1	82.59	50	242	+192
IEF	1	94.36	80	211	+131
VNQ	0	86.87	90	0	-90
DBC	0	21.43	60	0	-60
SHV	-	110.35	0	544	+544

3.4 Trade Execution with IBKR

The strategy can be executed through Interactive Brokers using two levels of automation:

1. A semi-automated method via TWS basket orders
2. A fully automated method using the IBKR API

The Python code provided in the notebook supports both approaches.

Semi/Automatic Execution via TWS Basket File

For investors who prefer a more hands-on approach, the notebook generates a .csv file containing a list of trade orders. Each row in the file specifies:

- The financial instrument (ETF)
- Action (Buy or Sell)

- Quantity (Shares)
- Order type (Market-On-Close, MOC)

This file can be uploaded directly into Interactive Brokers' Trader Workstation (TWS) using the BasketTrader tool. Figure 5 shows an example of the basket file loaded in TWS. Note that MOC orders must be transmitted to the exchange at least 15 minutes before the market closes (typically by 3:45 PM ET) to ensure execution.

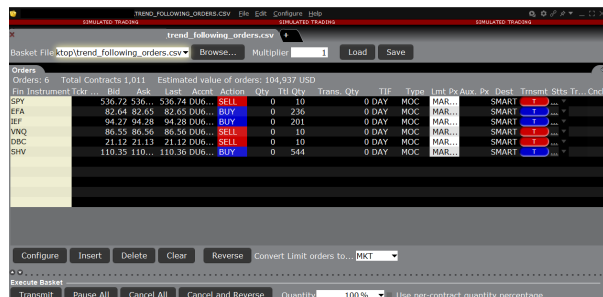


Figure 5: Trader Workstation showing a loaded .csv basket file with Market-On-Close (MOC) orders, ready for batch execution via the BasketTrader tool.

Automated Execution via IBKR API

For investors looking to fully automate the rebalancing process, the notebook integrates with Interactive Brokers through the `ib.async` package. This enables Python to connect directly to TWS or IB Gateway using a socket-based connection.

The automation process includes:

- Establishing a connection to IBKR
- Validating each contract to confirm it's tradable
- Submitting Market-On-Close (MOC) orders programmatically

Figure 6 illustrates the communication flow between the Jupyter notebook and Interactive Brokers. Once all contracts are validated, the notebook submits the MOC orders.

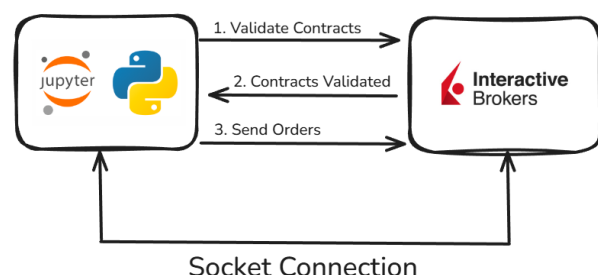


Figure 6: Diagram showing the API-based execution flow between the Python notebook and Interactive Brokers via socket connection.

Once all contracts have been validated and confirmed as tradable, the notebook proceeds to submit Market-On-Close (MOC) orders directly to Interactive Brokers. This process ensures that each trade aligns precisely with the portfolio targets calculated during the rebalancing phase.

Figure 7 illustrates the full execution workflow. The top portion of the figure shows the Jupyter Notebook confirming that each MOC order has been successfully sent to IBKR. These confirmations include key details such as the ticker symbol, quantity, action (Buy or Sell) and Order ID.

The bottom portion of the Figure 7 displays the same orders as they appear inside Trader Workstation (TWS), Interactive Brokers' professional trading platform. Each instruction submitted via Python is mirrored in TWS, ensuring full transparency and verifiability.

To further enhance clarity, color-coded arrows are used to visually match each order across both systems, from the notebook to the broker interface. This visual feedback confirms that the automated workflow performs as expected and that programmatic instructions result in accurate broker-side execution.

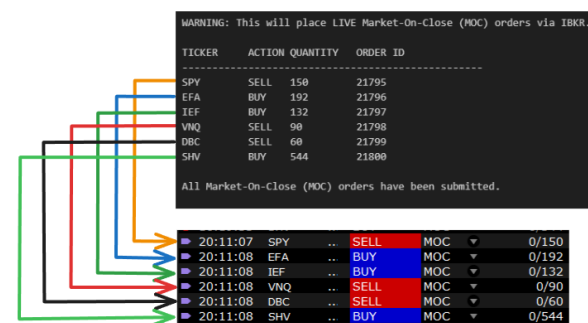


Figure 7: Confirmation of Market-On-Close (MOC) Order Execution Across Jupyter and Trader Workstation

4 Conclusion

This paper revisits Meb Faber's classic Global Tactical Asset Allocation (GTAA) model and provides both updated results and practical enhancements aimed at improving its real-world applicability.

First, we extended Faber's original analysis by evaluating the model over a 20-year *out-of-sample* period from 2006 to March 2025. Despite multiple market shocks, including the Global Financial Crisis, the COVID-19 pandemic, and inflation-driven volatility, the strategy delivered a Sharpe Ratio of 0.68, a compound annual growth rate (CAGR) of 6.05%, and a maximum drawdown of 11.7%. These figures

confirm the robustness of the original approach, even during challenging macroeconomic conditions.

Next, we identified a key implementation risk in the form of rebalance timing luck. Depending on the specific trading day used for signal evaluation and execution, performance outcomes varied by over 2% in CAGR. To address this, we introduced a weekly tranching rebalancing approach, in which the portfolio is split into four sub-portfolios, each rebalanced on a different week. This method reduced CAGR dispersion to 0.6% and cut annual turnover by nearly half compared with a weekly rebalanced implementation.

Finally, we focused on practical implementation. We presented a Python-based framework that enables investors to automate the GTAA strategy using Interactive Brokers. Two execution methods are supported: a semi-automated approach via TWS basket orders, and a fully automated system using the IBKR API. A complete Jupyter Notebook is available on Google Colab ([link⁸](#)) to facilitate replication and customization.

Taken together, our results suggest that the GTAA framework remains a viable and implementable strategy for investors seeking systematic exposure with built-in risk management. When enhanced with simple yet effective techniques, such as tranching rebalancing and API-based automation, it becomes an even more attractive solution for long-term portfolio construction.

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CONTACT US

Concretum SA
Viale Carlo Cattaneo 1
CH-6900 Lugano
Switzerland

info@concretumgroup.com
www.concretumgroup.com

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